

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Seasoned Senior Software Engineer with 10+ years of experience building **high-performance backend systems, low-latency APIs, and distributed architectures** across fintech, trading, and large-scale SaaS platforms. Adept at **Rust, C++, C#, and Go**, with strong expertise in **multithreaded systems, async programming, real-time data streaming, and high-throughput services**. Proven success in **optimizing latency-sensitive trading workflows, designing fault-tolerant execution engines, and integrating with global financial APIs**. Passionate about **high-frequency trading infrastructure, system performance, and crypto exchange connectivity**.

TECHNICAL SKILLS

- **Programming Languages:** Rust, C++, C#, Go, Java, Python, TypeScript, JavaScript
- **Frameworks & Tools:** .NET Core, Node.js, Kafka, gRPC, RabbitMQ, Redis, Docker, Kubernetes
- **Databases:** SQL Server, PostgreSQL, MySQL, MongoDB, Cassandra, Redis (SQL + NoSQL)
- **Cloud & Infrastructure:** AWS (EC2, S3, IAM, VPC, Lambda), GCP (Pub/Sub, BigQuery, GKE), Azure
- **Trading & Systems Expertise:** Low-latency design, HFT execution engines, market data ingestion, FIX/REST/WebSocket APIs, concurrent programming, capital allocation logic
- **DevOps & Monitoring:** Terraform, Helm, Prometheus, Grafana, ELK, CI/CD pipelines, GitLab CI, Azure DevOps
- **Other:** Distributed Systems, Event-Driven Architecture, Microservices, Security by Design

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Architected and implemented **low-latency backend services in Rust and C#** that reduced response times by 40% for high-throughput financial workflows, ensuring consistent sub-millisecond execution.
- Built **fault-tolerant real-time APIs and messaging pipelines with Kafka and gRPC** to handle streaming data ingestion exceeding 500K events per second in live trading and analytics platforms.
- Designed and optimized **multithreaded services with async programming models** to support concurrent order execution, lowering processing delays under peak market volatility conditions.

- Integrated trading engines with **multiple crypto exchanges (Binance, Coinbase, Kraken)** via FIX, REST, and WebSocket APIs, enabling high-volume automated trade routing and execution logic.
- Collaborated with traders and quant engineers to **transform market-making strategies into robust execution algorithms**, resulting in a measurable 25% increase in trading efficiency.
- Implemented **real-time monitoring, logging, and anomaly detection systems** with Prometheus and Grafana to ensure 99.99% uptime in mission-critical production environments.
- Drove **system-level performance tuning in memory management, CPU affinity, and network optimizations** that significantly improved throughput in high-frequency order matching.
- Mentored a team of engineers in **best practices for low-latency design, distributed architecture, and crypto-specific trading infrastructure**, ensuring knowledge transfer and code excellence.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Engineered **high-performance backend microservices** in .NET Core and C++ to process millions of real-time financial transactions securely across distributed infrastructures.
- Designed and maintained **low-latency APIs and capital allocation logic** for automated trading workflows, enabling scalable support for institutional-grade financial applications.
- Improved **throughput by 35% through optimized database indexing and concurrency handling** across SQL and NoSQL platforms supporting large data pipelines.
- Integrated with **external market APIs and financial platforms**, ensuring seamless cross-exchange order execution and settlement with high accuracy.
- Developed **fault-tolerant distributed services** leveraging Redis, RabbitMQ, and Kafka, ensuring resilience during network or infrastructure disruptions.
- Enhanced **crypto portfolio management systems with real-time analytics dashboards** that provided traders with latency-sensitive insights into execution performance.
- Collaborated on **system-level security enhancements**, implementing encryption, OAuth2 authentication, and secure key management for sensitive data flows.
- Contributed to **CI/CD pipelines and automated testing frameworks**, enabling rapid, high-confidence deployments in production trading environments.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Designed and developed **backend modules for trading and fintech applications**, focusing on concurrency, performance optimization, and reliability under high load.
- Implemented **multi-threaded services in C# and Java** to handle complex order routing workflows, ensuring fast response times during high-volume trading periods.
- Built **real-time monitoring services** that tracked latency, throughput, and execution metrics across distributed clusters, improving visibility into system health.
- Integrated **payment and settlement systems with SQL Server and PostgreSQL**, supporting high transaction volumes with ACID guarantees.
- Contributed to **low-latency data ingestion pipelines** from external APIs, enabling real-time streaming analytics and alerting for market events.
- Refactored legacy systems into **scalable microservices** deployed on Docker and Kubernetes, modernizing system design and reducing downtime.
- Participated in **cross-functional discussions with product teams and trading analysts**, ensuring business requirements translated into optimized technical solutions.
- Enhanced **system testing frameworks and simulation environments** to validate latency-sensitive execution logic before production rollout.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Developed backend services in **C# and Node.js** for digital asset management platforms with emphasis on scalability and resilience.
- Implemented **integration with third-party APIs and real-time event streams** for asset synchronization across distributed systems.
- Designed **concurrent background services with thread-safe execution models** to improve system throughput under peak demand.
- Improved **SQL database query performance** by optimizing indexes and stored procedures, reducing execution time across transactional workloads.
- Built **CI/CD workflows using Jenkins and Docker**, automating deployments and reducing release cycles by 50%.
- Contributed to **logging and monitoring frameworks** that tracked latency and execution health in staging and production environments.
- Collaborated closely with team leads to **define architecture for scalable backend modules** that aligned with long-term growth and performance goals.
- Delivered **robust unit and integration test coverage**, improving system reliability and confidence in critical backend components.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Assisted in developing **backend modules in C# and Java** for enterprise applications, focusing on concurrency and system-level performance tuning.
- Supported the **implementation of asynchronous APIs and background workers**, ensuring improved responsiveness under heavy user loads.
- Worked on **database design and query optimization** for SQL Server, enhancing transaction throughput for real-time applications.
- Built **prototype data ingestion pipelines from external APIs** to explore real-time integration opportunities in trading workflows.
- Contributed to **internal monitoring tools** that tracked API response latency and system health metrics across environments.
- Collaborated with senior engineers in **architecture discussions for distributed systems**, gaining early exposure to high-throughput system design.
- Enhanced **security mechanisms** in backend workflows by applying encryption and role-based access controls.
- Documented technical processes and created **internal knowledge-sharing guides**, supporting efficient onboarding of future interns and engineers.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ AWS Certified Solutions Architect – Associate
 - ❖ GCP Professional Cloud Developer
 - ❖ Microsoft Certified: Azure Solutions Architect Expert
 - ❖ Certified Kubernetes Administrator (CKA)
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PROJECTS

- ✓ **High-Frequency Crypto Trading Engine (2024):** Designed and built a **Rust-based execution engine** with sub-millisecond latency, integrated with Binance/Kraken APIs, and deployed on AWS EC2 with monitoring via Prometheus.
- ✓ **Capital Routing & Allocation Module (2023):** Developed **real-time capital allocation service** in C# and Go, managing liquidity across multiple exchanges and reducing idle capital by 22%.
- ✓ **Real-Time Analytics Dashboard (2022):** Built a **low-latency metrics dashboard** for traders with Kafka and WebSockets, providing real-time visibility into order execution, slippage, and market depth.